

Noise-Aware Ansatz Design for Digitized Adiabatic Quantum Computing

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We benchmark eight digitized-adiabatic ansatz variants for approximating the ground state of the transverse-field Ising model at $h = J$, $N = 4$ qubits on a Heron-r2-calibrated noisy density-matrix simulator. The variants combine three binary design choices—interpolation schedule, Trotter decomposition, and the presence or absence of a first-order counterdiabatic correction derived from the adiabatic gauge potential—and are optimized with COBYLA against the energy expectation value $\langle \hat{H} \rangle$ across depths $p \in \{1, \dots, 9\}$. The exact ground energy is $E_0 = -4.7588$. The best-performing variant, Linear + Suzuki–Trotter without counterdiabatic correction at $p = 4$, achieves $E^* = -4.645$, a residual of 0.114 above E_0 . The linear and trigonometric schedules produce essentially identical optimized energies for the non-counterdiabatic variants at all depths, but diverge by up to ~ 0.3 for the counterdiabatic variants past $p = 2$, indicating that schedule shape couples to the optimization landscape only when the additional variational freedom of CD is present. Suzuki–Trotter drops to within 0.05 of its eventual floor by $p = 2$ ($E = -4.603$ at $p = 2$, vs. $E = -4.119$ for Lie–Trotter at the same depth); the two orderings reach their absolute optima at $p^* = 4$ (Suzuki, $E = -4.645$) and $p^* = 5$ (Lie, $E = -4.625$), and converge to the same noise floor at large p . Counterdiabatic driving does not produce an algorithmic gain at any depth ≥ 3 and degrades monotonically past its optimum at $p^* = 2$. We identify the dominant cost: the threefold-larger entangling-gate count of the counterdiabatic layer admits proportionally more noise per layer, and the additional variational freedom of the CD ansatz lets the optimizer find low-energy states that drift away from the true ground state as depth grows. These results establish that a counterdiabatic correction which is exact in the closed-system limit can be a net loss on hardware-realistic noise once gate-count overhead and optimizer dynamics are accounted for.

INTRODUCTION

Adiabatic quantum computation prepares the ground state of a problem Hamiltonian $\hat{H}_1$ by initializing the system in the ground state of a driver $\hat{H}_0$ and slowly evolving a time-dependent Hamiltonian between the two. The standard interpolation is $\hat{H}(t) = (1 - t)\hat{H}_0 + t\hat{H}_1$. We instead use $\hat{H}(t) = \hat{H}_0 + t\hat{H}_1$, whose endpoint $\hat{H}(t = 1) = \hat{H}_0 + \hat{H}_1$ is exactly the full transverse-field Ising Hamiltonian whose ground state we want to prepare. The adiabatic theorem requires that the evolution time T scale as the inverse-square of the minimum energy gap along the path [1, 2].

Digitized adiabatic quantum computation realizes this continuous evolution on gate-based hardware by Trotterizing the time-ordered exponential into a finite sequence of single- and two-qubit gates. This is the central tension of the problem. The finite total time T contributes a non-adiabatic error $\epsilon_{\text{ad}} \sim 1/T$, while the finite step count n contributes a Trotter error that originates from the non-commutativity of $\hat{H}_0$ and $\hat{H}_1$. The Baker–Campbell–Hausdorff formula gives a leading per-step error $-\frac{1}{2}[\hat{H}_0, \hat{H}_1]\delta t^2$, and compounding over n steps with $\delta t = T/n$ yields $\epsilon_{\text{Tr}} \sim T^2/n$ for first-order Lie–Trotter and $\epsilon_{\text{Tr}} \sim T^3/n^2$ for second-order Suzuki, where the leading commutator term cancels by symmetry and the residual is set by nested commutators [3].

On noisy intermediate-scale quantum (NISQ) hardware, gate noise dominates over both of these error sources. It grows with circuit depth and is largely set by the two-qubit gate fidelity. For IBM’s Heron r2 architecture the median two-qubit ECR gate error is approximately 1.95×10^{-3} , so a circuit with ~ 100 entangling gates has lost most of its fidelity to gate noise. In this regime, an ansatz design choice that improves

algorithmic accuracy by a constant factor but adds entangling structure is likely a net loss.

In this work, we ask which ansatz designs make best use of a fixed depth budget under realistic gate noise. We treat the digitized adiabatic protocol as a variational ansatz where the Trotter angles are optimized against the energy expectation value — structurally identical to the quantum approximate optimization algorithm (QAOA) [4], applied here to ground-state preparation rather than combinatorial optimization, following the digitized adiabatic approach of Barends et al. [5]. We systematically compare three design dimensions across all eight combinations, evaluated on a Heron-r2-calibrated noisy density-matrix simulator at $N = 4$ qubits across depths $p \in \{1, \dots, 9\}$.

We had originally intended to execute the depth- p^* optimized circuits on `ibm_kingston` (Heron-r2), but exhausted our IBM Open-Plan allocation before completing a full design-point sweep with adequate shot statistics. The data reported here is therefore restricted to the noisy simulator. Hardware execution is left to future work.

PROBLEM STATEMENT AND ANSATZ DESIGN

Model

We target the ground state of the open-chain transverse-field Ising model

$$\hat{H} = -J \sum_{i=1}^{N-1} \hat{Z}_i \hat{Z}_{i+1} - h \sum_{i=1}^N \hat{X}_i, \quad (1)$$

at the critical-point parameter $h = J = 1$, with $N = 4$ qubits throughout. Setting $h = J$ places the system at its quantum critical region, where the true ground state is highly entangled and any product-state ansatz is provably suboptimal.

For the digitized adiabatic protocol, we identify the problem and driver pieces of $\hat{H}$ as

$$\hat{H}_1 = -J \sum_{i=1}^{N-1} \hat{Z}_i \hat{Z}_{i+1}, \quad \hat{H}_0 = -B_x \sum_{i=1}^N \hat{X}_i, \quad (2)$$

with $B_x = h = 1$ at the endpoint of the schedule. The driver's ground state $|+\rangle^{\otimes N}$ is prepared by applying a layer of Hadamard gates. At $N = 4$ and $h = J = 1$, classical diagonalization of $\hat{H}$ gives the exact ground energy $E_0 = -4.7588$.

At depth p , each ansatz applies a Trotter unitary

$$U(\gamma, \beta) = \prod_{k=1}^p U_1(\gamma_k) U_0(\beta_k) \quad (3)$$

to $|+\rangle^{\otimes N}$, where U_1 and U_0 implement evolution under $\hat{H}_1$ and $\hat{H}_0$ respectively, and $\gamma, \beta \in \mathbb{R}^p$ are classically optimized.

Schedule

The variational angles are seeded from a midpoint-sampled schedule, where $\tau_k = (k - \frac{1}{2})/p \in [0, 1]$ denotes the k -th layer's normalized position. We seed $\gamma_k = (T/p) t(\tau_k)$ for the problem and $\beta_k = T/p$ for the driver, comparing a linear schedule $t(\tau) = \tau$ and a phase-shifted trigonometric schedule $t(\tau) = \tau + (c/2\pi) \sin(2\pi\tau + \pi/2)$ with $c = 0.3$, both passed to COBYLA as starting points for energy minimization. The phase shift is chosen so that the trigonometric perturbation does not vanish at the midpoint sample $\tau = 0.5$, which would otherwise make the linear and trigonometric seeds coincide at $p = 1$ and propagate that coincidence through our warm-start chain.

Trotter Ordering

Lie–Trotter (first-order) decomposition applies one U_0 followed by one U_1 per layer. Suzuki–Trotter (second-order) decomposition applies $U_0^{1/2} U_1 U_0^{1/2}$, doubling the number of single-qubit U_0 layers (with halved angles) but leaving the entangling U_1 count unchanged. The local Trotter error drops from $\mathcal{O}(\delta t^2)$ to $\mathcal{O}(\delta t^3)$.

Counterdiabatic Driving

A first-order counterdiabatic (CD) term derived from the adiabatic gauge potential adds a nearest-neighbor correction

$$U_{CD}(\alpha_k) = \exp \left[-i\alpha_k \sum_{i=1}^{N-1} (\hat{Y}_i \hat{Z}_{i+1} + \hat{Z}_i \hat{Y}_{i+1}) \right] \quad (4)$$

to each layer of the Trotter circuit, with α_k a free variational parameter alongside γ_k and β_k . The structure of the CD term comes from the leading-order solution of the adiabatic gauge potential $\hat{A}_\lambda$, which generates the unitary that exactly cancels diabatic transitions during a finite-time schedule. For our choices of $\hat{H}_0$ and $\hat{H}_1$, the leading commutator $[\hat{H}_0, \hat{H}_1] = -2BJ \sum_i (\hat{Y}_i \hat{Z}_{i+1} + \hat{Z}_i \hat{Y}_{i+1})$ generates the dominant non-adiabatic drift, and the variational ansatz $\hat{A}_\lambda \propto \sum_i (\hat{Y}_i \hat{Z}_{i+1} + \hat{Z}_i \hat{Y}_{i+1})$ is the simplest local gauge-potential correction that cancels it [6]. CD driving cancels leading-order non-adiabatic transitions and, in noiseless simulation, lets a shallower circuit reach the same ground-state fidelity as a deeper non-CD ansatz; equivalently, it relaxes the adiabatic theorem's evolution-time requirement [6].

The cost is in the gate count. Although the two CD operators on each bond commute, the per-bond exponential has no native single-gate decomposition on the heron r2 architecture. It is implemented as two RZZ rotations conjugated by $RX(\pm\pi/2)$ basis changes per bond per layer, raising the per-layer entangling-gate count from 3 to 9 at $N = 4$. Whether this overhead is offset by the algorithmic gain in fidelity is the question the depth sweep is designed to answer.

The three binary choices give a $2^3 = 8$ design space. We benchmark all eight combinations and treat the simplest—linear schedule, Lie–Trotter, no CD—as the baseline.

METHODOLOGY

Energy Evaluation

We evaluate the energy expectation value as $\langle \hat{H} \rangle = \text{Tr}(\rho \hat{H})$, building $\hat{H}$ as a full $2^N \times 2^N$ matrix that includes both the $\hat{Z}\hat{Z}$ couplings and the transverse-field $\hat{X}$ term. The direct trace evaluation is exact and cheap at $N = 4$. We use this evaluation at every COBYLA iteration.

Optimization Pipeline

For each design and depth $p \in \{1, \dots, 9\}$, the variational parameters are initialized from the schedule values described in Section II and optimized with COBYLA against $\langle \hat{H} \rangle$ measured on the noisy density-matrix simulator; CD parameters α_k are seeded at $\alpha_0 = 0.02$. We warm-start each depth from the converged parameters of depth $p - 1$, linearly interpolated onto p samples. The COBYLA budget is 100 iterations per variational parameter with a floor of 500. We report both the energy $\langle \hat{H} \rangle$ and the fidelity $\mathcal{F}(\rho) = \langle \psi_0 | \rho | \psi_0 \rangle$: energy captures leakage into excited states weighted by their energy distance from the ground state, while fidelity isolates the overlap with $|\psi_0\rangle$ alone. The total evolution time is fixed at $T = 7$, the smallest integer at which a deep ($p = 40$) noiseless Suzuki–Trotter circuit reaches within 0.5% of $|E_0|$ under our seeded schedule.

Heron-r2 Noise Model

We use a noise profile drawn from `ibm.kingston` (Heron-r2): single-qubit gate error 2×10^{-4} , two-qubit gate error 1.95×10^{-3} , readout error 9.22×10^{-3} , $T_1 = 280 \mu\text{s}$, $T_2 = 154 \mu\text{s}$, single-qubit gate duration 35 ns, two-qubit gate duration 80 ns, and measurement duration 1200 ns [7]. The depolarizing channel and the thermal-relaxation channel are composed per gate; readout errors are applied symmetrically as a per-qubit bit-flip probability.

RESULTS

Depth Sweep

Figure 1 shows the energy and fidelity reached by each of the eight ansatz variants on the noisy density-matrix simulator across depths $p \in \{1, \dots, 9\}$.

A clean separation emerges between the four non-counterdiabatic variants and the four counterdiabatic ones. The non-counterdiabatic variants improve sharply from $p = 1$ to $p \approx 4$, where the energy reaches $E \approx -4.62$ and fidelity rises above 0.95. The Suzuki–Trotter variants drop to within 0.05 of their eventual optimum by $p = 2$ ($E = -4.603$) while Lie–Trotter requires $p = 4$ to reach the same band; the two orderings achieve their absolute optima at $p^* = 4$ (Linear + Suzuki, $E = -4.645$) and $p^* = 5$ (Trigonometric + Lie, $E = -4.625$) respectively, with Suzuki marginally lower. Beyond the per-variant optimum, both metrics drift mildly in the wrong direction as cumulative gate noise overtakes the per-layer Trotter improvement: by $p = 9$, all four non-counterdiabatic variants have settled at $E \approx -4.54$ with fidelity ≈ 0.94 .

The four counterdiabatic variants follow more erratic trajectories. The Lie–Trotter + CD variants start at $E \approx -0.19$ at $p = 1$, drop sharply to $E = -4.541$ at $p = 2$, then degrade through depth, with the trigonometric variant settling at $E = -4.091$ and the linear variant at $E = -4.016$ by $p = 9$. The Suzuki–Trotter + CD variants reach $E \approx -4.532$ at $p = 2$ but are non-monotonic past their optimum, with Linear ending at $E = -4.042$ and Trigonometric at $E = -3.999$ by $p = 9$. None of the counterdiabatic variants match the best non-counterdiabatic energy at any depth beyond $p = 2$.

The schedule comparison is more informative than the original test we proposed. For the non-counterdiabatic variants, the linear and trigonometric schedules produce optimized energies that agree to within ~ 0.005 at every depth, indicating that COBYLA’s basin of attraction is large enough to absorb the schedule-induced seed differences. For the counterdiabatic variants, however, the schedules diverge substantially past $p = 2$ — by up to ~ 0.3 in energy at intermediate depths and persisting to $p = 9$. We attribute this to the additional variational freedom of the CD ansatz: the third per-layer parameter α_k amplifies the influence of the seed direction on which

local minimum COBYLA reaches, so the schedule shape couples non-trivially to the optimization landscape only when CD parameters are present. The Lie + CD trajectory benefits from the trigonometric schedule at most depths past $p = 2$; the Suzuki + CD trajectory benefits from the linear schedule at most of the same depths.

Variant Comparison at the Optimal Depth

TABLE I. Optimized energy at the depth p^* that minimizes simulator energy, for all eight ansatz variants. $N = 4$, $\hbar = J = 1$, exact ground energy $E_0 = -4.7588$. The linear and trigonometric variants agree closely for the non-counterdiabatic variants but diverge for the counterdiabatic ones, indicating that the additional variational freedom of CD couples non-trivially to the seed direction.

Variant	p^*	E^*	$E^* - E_0$
Linear + Lie-Trotter	5	-4.624	+0.135
Linear + Lie-Trotter + CD	2	-4.541	+0.218
Linear + Suzuki-Trotter	4	-4.645	+0.114
Linear + Suzuki-Trotter + CD	2	-4.532	+0.227
Trigonometric + Lie-Trotter	5	-4.625	+0.134
Trigonometric + Lie-Trotter + CD	2	-4.541	+0.218
Trigonometric + Suzuki-Trotter	5	-4.621	+0.138
Trigonometric + Suzuki-Trotter + CD	2	-4.532	+0.227

Trotter Ordering Effects within the Non-Counterdiabatic Variants

Suzuki–Trotter and Lie–Trotter share entangling-gate counts at the same p , so any difference between them is attributable to the second-order Trotter cancellation of the leading commutator term. We see this most clearly at $p = 2$: Suzuki achieves $E = -4.603$ while Lie reaches only $E = -4.119$, a gap of 0.48. By $p = 4$ both orderings have closed this gap (Suzuki $p^* = 4$ gives $E = -4.645$; Lie at $p = 4$ gives $E = -4.596$, with Lie continuing to improve to $p = 5$). The two orderings reach their absolute optima at adjacent depths ($p^* = 4$ vs. $p^* = 5$) with Suzuki marginally lower (-4.645 vs. -4.625). At larger p both converge to the same noise floor near $E \approx -4.54$ at $p = 9$, consistent with both having driven Trotter error well below the noise contribution.

Counterdiabatic Variants

The counterdiabatic variants peak early and degrade with depth. Both Lie + CD and Suzuki + CD reach their optima at $p^* = 2$, with $E = -4.541$ and $E = -4.532$ respectively — both substantially worse than the corresponding non-CD optima. Past $p^* = 2$, all four CD variants degrade non-monotonically, with the trajectory shape depending on schedule choice (described above) but the overall trend toward higher energies common to all of them. By $p = 9$ the CD vari-

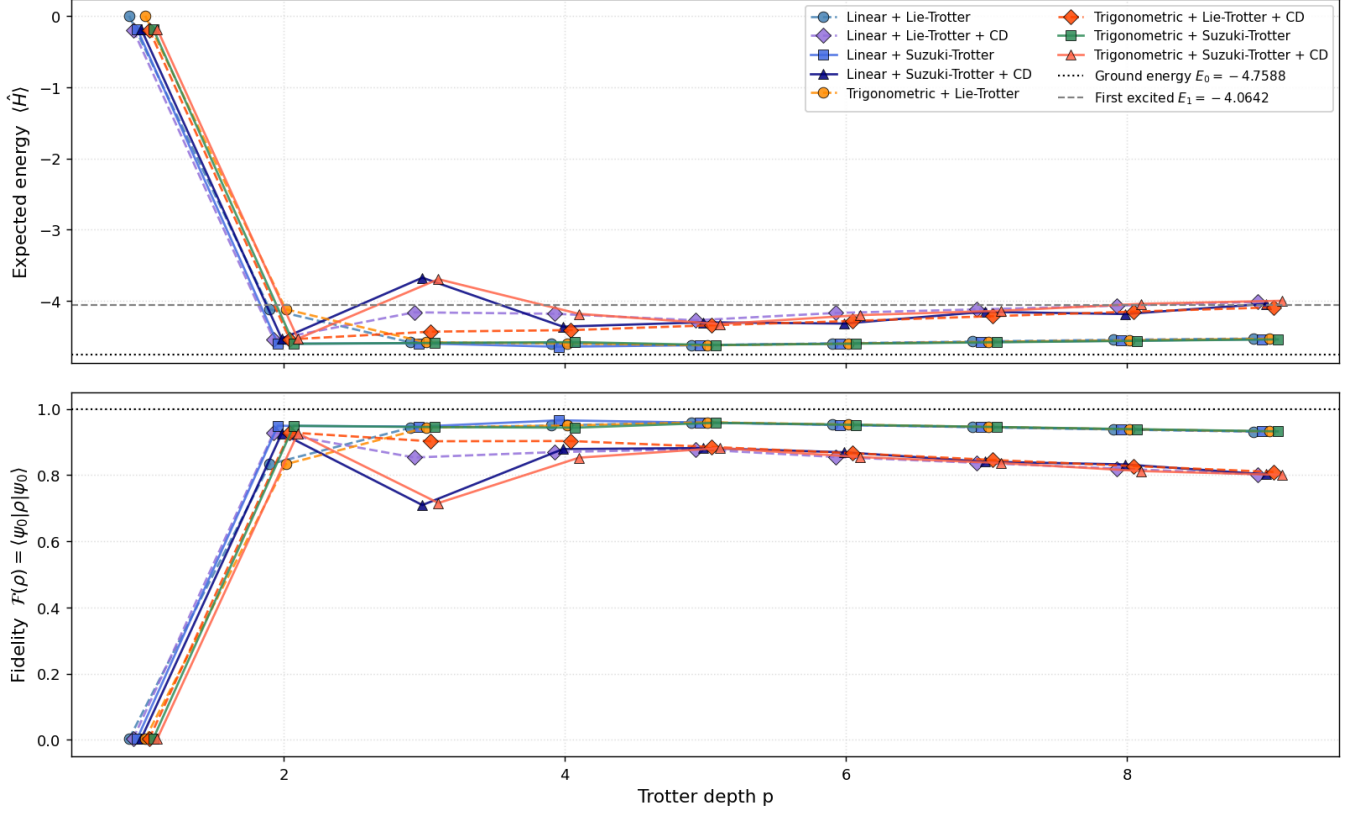


FIG. 1. Depth sweep at $N = 4$, $h = J = 1$, on the Heron-r2-calibrated noisy density-matrix simulator. *Top*: optimized energy expectation value $\langle \hat{H} \rangle$. Dotted line: exact ground energy $E_0 = -4.7588$. Dashed line: first excited energy $E_1 = -4.0642$. *Bottom*: fidelity to the true ground state, $\mathcal{F}(\rho) = \langle \psi_0 | \rho | \psi_0 \rangle$. Dotted line: perfect fidelity.

ants have settled at $E \in [-4.09, -4.02]$, well above the non-CD plateau near $E \approx -4.54$. The strict average degradation past p^* confirms the gate-count cost of CD is not offset by an algorithmic gain on this noise model.

Notably, all four CD variants converge toward $E_1 = -4.0642$ at deep p rather than toward the noise-mixed plateau of the non-CD variants near $E \approx -4.54$. All four CD endpoints at $p = 9$ fall within 0.07 of E_1 , with three of the four within 0.05. Whatever mechanism drives the CD degradation, the converged states are not generic noise-mixed states but are concentrated near the first excited eigenenergy of $\hat{H}$. We do not commit to a specific mechanism for this convergence here, but note that it suggests the CD optimizer is finding a structured family of low-energy states rather than drifting at random, and that any future analysis of CD failure modes on NISQ hardware should treat the CD-versus-non-CD difference as a sector-level effect rather than a continuous degradation.

DISCUSSION

Trotter-Limited and Noise-Limited Regimes

The non-counterdiabatic curves cleanly separate two regimes. From $p = 1$ to the per-variant optimum ($p^* = 4$ for Suzuki, $p^* = 5$ for Lie), the energy and fidelity improve sharply: each new Trotter layer reduces $\delta t = T/p$ and brings the digitized circuit closer to the continuous adiabatic evolution. Beyond p^* , the curves reverse direction and drift away from the ground state at roughly 0.01 in energy per layer. This is the noise-limited regime: each additional Trotter layer adds three RZZ rotations and four RX rotations whose noise contribution outpaces the residual Trotter improvement.

The size of the noise penalty at the optimum is ~ 0.11 in energy ($E^* - E_0 \approx 0.114$ for Linear + Suzuki at $p^* = 4$). This sets the floor of what the variational protocol can achieve on this noise model: gate noise prevents the optimizer from reaching closer than $\sim 2.4\%$ of the true ground energy regardless of the design choices we explored.

Suzuki Beats Lie Where Algorithmic Error Dominates

Suzuki–Trotter and Lie–Trotter share entangling-gate counts at the same p , so any difference between them is attributable to the second-order Trotter cancellation of the leading commutator term. We see two effects: Suzuki drops sharply earlier than Lie (at $p = 2$ Suzuki achieves $E = -4.603$ while Lie reaches only $E = -4.119$), and its absolute floor is marginally lower (-4.645 at $p^* = 4$ vs. -4.625 at $p^* = 5$). The earlier drop is consistent with the second-order $\mathcal{O}(\delta t^3)$ Trotter error scaling: Suzuki at $p = 2$ has the same $\delta t = 3.5$ as Lie at $p = 2$, but its smaller per-step error means the Trotter contribution at $p = 2$ is already much smaller than the first-order penalty Lie pays at the same depth. By $p = 4$ both orderings have driven Trotter error below the noise floor and the residual difference between them is marginal.

Counterdiabatic Driving and Gate-Count Overhead

The most informative finding in our data is that the counterdiabatic correction does not produce an algorithmic gain at any depth ≥ 3 on this noise model, and produces a depth-monotonic energy degradation past p^* . We identify the dominant cost as gate-count overhead.

The CD term adds two RZZ rotations and four RX rotations per nearest-neighbor pair per layer, raising the per-layer entangling-gate count from 3 to 9 at $N = 4$. On a per-layer basis, the gate-noise penalty of a counterdiabatic layer is roughly three times that of a bare Trotter layer. This alone is sufficient to push p^* down to 2 for the CD variants and to make every CD layer past that net-negative on average.

A secondary effect is the increased variational freedom of the CD ansatz. With three parameters per layer (vs. two for the bare Trotter ansatz), the CD optimization landscape admits a wider family of low-energy states that COBYLA can converge onto, including states whose energy expectation value is low but whose overlap with the true ground state is poor. This is most visible in the schedule-dependence of the CD variants past $p = 2$: with two extra degrees of freedom interacting with the warm-start chain, the seeded direction can carry COBYLA toward different local minima for linear vs. trigonometric schedules. The non-CD variants do not show this sensitivity, consistent with their smaller-dimensional parameter space being absorbed by COBYLA’s basin of attraction.

The methodological consequence is that NISQ-era ansatz design cannot be evaluated by algorithmic-error scaling alone. A counterdiabatic correction that is exact in the closed-system limit can still be a net loss on hardware-realistic noise once the gate-count overhead and the optimizer’s behavior under the larger parameter space are accounted for. Symmetry-respecting variants of the gauge-potential ansatz, ansätze co-designed for the native two-qubit gate set, qubit-error-aware angle scheduling, and hardware execution to compare these simulator-side trends against an actual device are the natural

next directions.

CONCLUSION

We benchmarked all eight combinations of three digitized adiabatic ansatz design choices—schedule, Trotter decomposition, and counterdiabatic driving—on the transverse-field Ising model at the critical-point parameter $h = J$, $N = 4$ qubits, on a Heron-r2-calibrated noisy density-matrix simulator. Four takeaways summarize the experiment.

(i) The optimal depth on this noise model is small: $p^* = 4$ for Suzuki–Trotter and $p^* = 5$ for Lie–Trotter (without CD), and $p^* = 2$ for both orderings with CD. Below p^* , accuracy is gated by Trotter error; above it, by gate noise. The best variant we tested (Linear + Suzuki–Trotter, no CD, $p = 4$) reaches $E^* = -4.645$ against the exact ground energy $E_0 = -4.7588$ —a residual of 0.114 that sets the noise floor of what this protocol can achieve.

(ii) Schedule shape is irrelevant for the non-counterdiabatic variants but matters for the counterdiabatic ones. Linear and trigonometric seeds produce optimized energies that agree to within ~ 0.005 for the non-CD variants at every depth, but diverge by up to ~ 0.3 for the CD variants past $p = 2$. We attribute this to the additional variational freedom of the CD ansatz coupling the seed direction to which local minimum COBYLA reaches.

(iii) Suzuki–Trotter beats Lie–Trotter at small p where algorithmic error dominates: at $p = 2$, Suzuki achieves $E = -4.603$ while Lie reaches only $E = -4.119$. The two orderings reach their absolute optima at adjacent depths ($p^* = 4$ vs. $p^* = 5$) with Suzuki marginally lower (-4.645 vs. -4.625), and converge to the same noise floor at large p .

(iv) Counterdiabatic driving does not produce an algorithmic gain at any depth ≥ 3 on this noise model, and degrades monotonically past its optimum at $p^* = 2$. The dominant cost is the threefold-larger entangling-gate count of the CD layer, which makes every CD layer past p^* net-negative on average. A secondary cost is the increased variational freedom of the CD ansatz, which lets the optimizer find schedule-dependent low-energy states that drift away from the true ground state as depth grows.

The methodological consequence is that NISQ-era ansatz design cannot be evaluated by algorithmic-error units or gate-count budgets alone. A counterdiabatic correction that is exact in the closed-system limit can still be a net loss on hardware-realistic noise once gate-count overhead and optimizer dynamics in the larger parameter space are accounted for. This points to productive directions for future work: symmetry-respecting variants of the gauge-potential ansatz, ansätze co-designed for the native two-qubit gate set, qubit-error-aware angle scheduling, and hardware execution to compare the simulator-side trends against an actual device.

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